

ADAM D. RUSHDY

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EDUCATION

Purdue University - B.S. Aeronautical & Astronautical Engineering, Honors

Expected Graduation: May 2028

GPA: 3.82/4.00

Activities: Purdue Aerial Robotics Team, Teaching Assistant, Purdue All-American Marching Band

WORK EXPERIENCE

Engineering Intern - *Xchanger*

Hopkins, MN | May - Aug 2026

- Diagnosed and troubleshot a production model error causing predicted pressure-drop to underestimate actual values by 2-3x, preventing inaccurate data sheets from reaching clients.
- Tested pressure-drop and velocity performance across 18 air-to-air heat exchanger units (100% of models) and multiple flow configurations, generating an empirical dataset to recalibrate the company's C-based modeling tool.
- Derived unit-specific correction factors now built into the production model, giving engineers more accurate datasheets and a validated safety margin when quoting new units.

Teaching Assistant - *Honors Intro to Engineering (ENGR 16X)*

West Lafayette, IN | Aug 2025 - Present

- Support first-year students with coursework, including 6 design projects covering modeling in Python, physics simulation in MATLAB, and robot design using Raspberry Pi embedded systems.

Intern - *Seminar for Top Engineering Prospects (STEP)*

West Lafayette, IN | Jun - Aug 2025

- Developed and delivered a modular Arduino programming curriculum to 320 prospective engineers, creating reusable functions and tutorials to teach fundamental microcontroller programming concepts.

Teaching Assistant - *Intro to C Programming*

West Lafayette, IN | Jan - May 2025

- Led 30+ students across 12 labs covering core C concepts, held office hours, and graded coding assignments.

TECHNICAL PROJECTS

R&D Airframe Member - *Purdue Aerial Robotics Team (PART)*

Sep 2025 - Present

- Performed structural/stress analysis in MATLAB to evaluate rib spacing trade-offs between panel buckling and structural weight margins, informing airframe design decisions.
- Developed CAM toolpaths for CNC machining of 2 precision carbon-fiber layup molds for wings within a 2-hour manufacturing constraint using Fusion 360. Fabricated composite wing assemblies using carbon fiber layup over laser-cut internal structures.
- Designed high-speed (90 m/s) UAV wing planform in XFLR5 and analyzed taper and twist effects on drag and lift distribution. Developing ANSYS Fluent CFD models of the wing, tail, fuselage, and full assembly to improve airframe geometry for low-drag.

UAV Inertial Navigation System

June - July 2026

- Built an INS using an ESP32 for real-time IMU, magnetometer, and GPS acquisition. In Python, implemented complementary filter attitude estimation and linear Kalman filtering for position/velocity estimation at 40 Hz.
- Reduced position RMSE from 13 m to 4.9 m (62%) by diagnosing filter lag during high-acceleration maneuvers and tuning Kalman process/measurement noise covariances.
- Validated estimator performance with post-flight replay against ArduPilot EKF output.

Fixed-Wing Surveillance UAV - Design, Analysis & Fabrication

Mar - June 2026

- Designed and fabricated a 1.3 m, 1.8 kg fixed-wing UAV with Raspberry Pi camera integration. Selected airfoils in XFOIL, sized aircraft weight and geometry for 25 m/s cruise, and modeled the full assembly in SolidWorks.
- Designed wing structure for a 6-g limit load factor, used FEA to confirm calculations & structural integrity with a maximum tip deflection of 12.1 mm. Fabricated using carbon-fiber skin layups over 3D-printed ribs and spars.
- Performed aerodynamic, stability, and flight dynamics analysis in XFLR5, iterating on CG, tail moment arm, and wing geometry to achieve a 6.67% static margin.
- Implemented ArduPilot flight controller for fly-by-wire and data logging. Completed maiden flight capturing airspeed, load factor, battery performance, and estimated position across a 150-second flight, validating the ~25 m/s cruise speed and maximum 3-g load against design predictions.

SKILLS

Programming: MATLAB/Simulink, Python, C, C++, Java, Git

CAD / Analysis: SolidWorks (CAD/FEA), Fusion 360 (CAD/CAM), NX, Teamcenter, XFLR5, XFOIL

Fabrication: Carbon-fiber composite layup, 3D printing (FDM), CNC milling

Leadership: Eagle Scout, Assistant Snare Segment Leader (Purdue AAMB)